Recognizable extensions: complete manuscript costs

The source is Antonio Díaz Ramos, Rémi Molinier and Antonio Viruel, *Every finite group is represented by a finite incidence geometry*, arXiv:2511.02410v2 (16 December 2025), thirteen pages. The accepted principal contract contains six source blocks and thirteen current proof contexts. It preserves the stated corrections and exclusions; it does not claim recovery of every historical auxiliary assertion.

Written papers. The primary ruler is calibrated NFC characters without whitespace. Titles, abstracts, body, references, author information and extractable figure labels remain counted. Only exact running furniture is removed.

Complete document	Pages	Characters
Source F010	13	25348
Common theory U_R	5	11288
Application R1	5	13688
Complete validation appendix	4	9025

The combined comparison pays for the whole common paper and application:

$$R_{\text{written}} = \frac{25348}{11288 + 13688} = \frac{25348}{24976} = 1.014894.$$

After acquiring the whole common paper and both complete validation programs, the conditional source/application comparison is

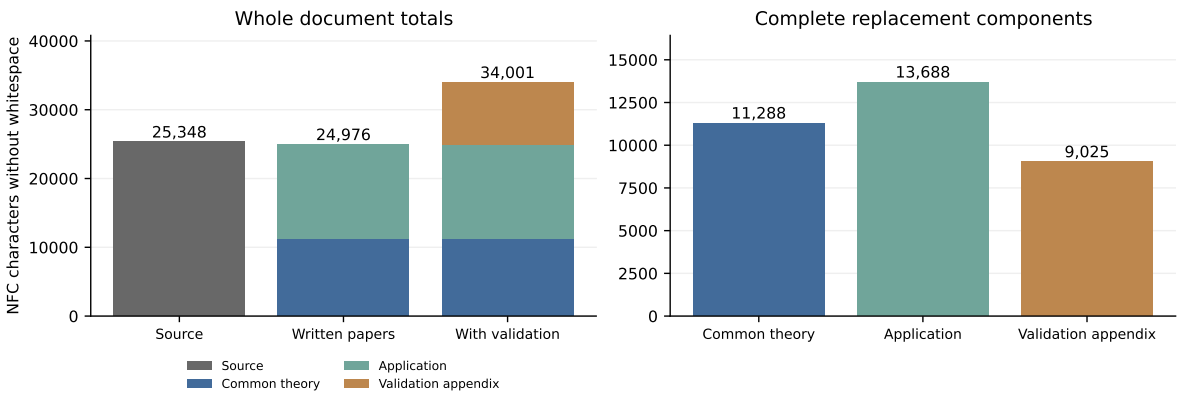
$$r_1 \mid (U_R, P_R) = \frac{25348}{13688} = 1.851841.$$

The conditional factor does not discard the acquisition cost. Other topics pay their own common theory in full, even where it overlaps.

Including the validation programs. Neither program is a premise of the uniform written proof. Both remain complete in the separate comparison. With the printed appendix,

$$R_{\text{printed}} = \frac{25348}{24976 + 9025} = \frac{25348}{34001} = 0.745507.$$

The two whole raw files instead cost 8310 characters, giving $25348/33286 = 0.761521$. These representations are alternatives, never simultaneous charges. Two complete historical drivers cost 13275 characters separately. Two full Lean files cost 5487 characters separately and contain eight generic and four reused local declarations; they do not formalize the complete application.



Both axes start at zero and every bar charges its complete documents. The written difference in this ruler is only 372 characters; the replacement including validation is longer than the source. A single application supplies no evidence of a recurring sawtooth pattern.

Sensitivity. For calibrated selected text in NFC:

Unit	Written	Conditional	With appendix
Whitespace words	1.008	1.831	0.896
Alphanumeric units	1.138	2.068	0.862
Nonspace characters	1.015	1.852	0.746
Characters, spaces collapsed	1.014	1.848	0.768
UTF-8 bytes, spaces collapsed	0.971	1.779	0.749

The UTF-8 written factor is below one: the small written saving is not stable across these units. The full JSON and CSV retain all five units under NFC and NFKC, each for untrimmed raw, selected raw and calibrated selected text: thirty coordinates and 120 exact fractions. Printed hyphens, code continuation marks and actual whitespace are preserved.

Scope and retained costs. The current proofs correct the chamber degrees to $r-1$ and $d(v)(r-1)$, the completed hexagon pair to (D_{12}, S_3), and the symmetric-family minimum-degree statement to $n \geq 3$. There is no universal rank-two representation or blanket $(S_n, A_n) = (\mathrm{Aut}(A_n), \mathrm{Inn}(A_n))$ claim. The original Theorem A already answers Question 2.4; no new question answer is claimed here.

The accepted independent input audit checked all 27 pages, 120 PDF metric cells, sixty separate raw-file cells and 566 glyph classes in 26 actual font dictionaries. Its 134 visible-glyph repairs include three byte-9 right braces; actual whitespace is otherwise preserved. The four mapsto strokes remain separate from adjacent arrows. The ruler uses pypdf 6.13.0 and FontTools 4.63.0. Source vector figures, edges, colors and formula placement are not fully represented by extracted text. Their cost, external proofs, historical software and operational engineering remain unknown, not zero. These factors describe the declared text representation, not total research effort or minimum proof complexity. Independent numerical review is required before publication.